Rahul Jha

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Professional Summary

Final Year B.Tech CSE student with a CGPA of 8.13/10. Hands-on experience in Python, SQL, Machine Learning and GenAI. Worked on hybrid RAG pipeline (Qdrant, BM25, cross encoder reranking), integration of LLM API(Gemini), Regression based price prediction and computer vision using YOLO. Expert in SQL (solved LeetCode SQL questions – 50 and worked with analytical SQL at AegisAI) and Databricks Lakehouse Fundamentals.

Education

SRM Institute of Science and Technology **Chennai, India**
B.Tech, Computer Science and Engineering – CGPA: 8.13/10.0 *2023 – 2027*

Experience

Software Engineer Intern **Jun 2026 – Jul 2026**
Siddharth Infosys *Remote*

Project: AAVA – AI Agricultural Voice Assistant | GitHub | Credentials

- Developed a Flask REST API backend for a multilingual voice assistant (Hindi, Marathi, English) providing farmers with up-to-date mandi prices and weather information.
- Developed an algorithm that predicts future trends using the least-square method of regression applied to time-series data from Agmarknet prices.
- Cut down time taken by Agmarknet price search from about 10 seconds to 0.4 seconds through state filter and header update; implemented API timeout and limit fallbacks.
- Integrated Agmarknet, OpenWeatherMap, and Gemini 2.0 Flash APIs; developed a browser voice-based interface using the Web Speech API and gTTS.
- **Tech:** Python, Flask, REST APIs, Gemini API, Linear Regression, Agmarknet API, OpenWeatherMap API, gTTS, JavaScript

Projects

AegisAI – Multimodal Incident Investigation Platform | GitHub | Live API | Live Dashboard

- Created a Python framework that analyzes events via FastAPI, PostgreSQL, SQL, and LangGraph, enabling structured analysis of the event and guiding tools to agents.
- Developed a hybrid RAG pipeline using Qdrant, BM25, Reciprocal Rank Fusion (RRF), and cross-encoder re-ranking, achieving 100% Precision@3 across 4 evaluation queries.
- Built machine learning/data processing pipelines based on YOLOv8 and ByteTrack using async processing through Celery/Redis with automated evaluation and GitHub Actions CI.
- **Tech:** Python, SQL, PostgreSQL, FastAPI, LangGraph, RAG, Qdrant, BM25, RRF, Cross-Encoder, YOLOv8, ByteTrack, Celery, Redis, GitHub Actions, Docker

Smart Surveillance & Incident Analysis System | GitHub | Live Dashboard

- Designed a full-stack ML application in Python/SQL to perform real-time object detection and tracking using FastAPI, PostgreSQL, YOLO26, and OpenCV.
- Used semantic search for incidents via Sentence Transformers and ChromaDB to create embeddings and retrieve incident information similar to the given input.
- Created and validated scalable data services using FastAPI, PostgreSQL, Docker, pytest, and GitHub Actions, employing data analysis and machine learning approaches.
- **Tech:** Python, SQL, PostgreSQL, MySQL, JavaScript, C++, C, HTML/CSS, PyTorch, YOLO26, OpenCV, ByteTrack, sentence-transformers, ChromaDB, FastAPI, React, Node.js, TypeScript, Docker, GitHub Actions, pytest

Technical Skills

Languages: Python, SQL, JavaScript, C++, C, HTML/CSS
SQL & Databases: PostgreSQL, MySQL, MongoDB, Redis, Neo4j; joins, aggregations, window functions, indexing, Database Design
Data Science & ML: PyTorch, YOLO, OpenCV, Linear Regression, Model Evaluation (precision@k, R^2)
GenAI: RAG, LLM APIs (Gemini), Prompt Engineering, LangGraph, MCP, sentence-transformers, ChromaDB, Qdrant
Cloud & Data Platforms: Databricks (Lakehouse Fundamentals), Render
Backend & Frontend: FastAPI, Flask, Node.js, Express.js, REST APIs (OpenAPI/Swagger), JWT, bcrypt, Celery, React.js, Next.js
Tools & DevOps: Git/GitHub, Docker, GitHub Actions, CI/CD, Linux, pytest
Core CS: Data Structures & Algorithms, OOP, DBMS, Computer Networks

Certifications & Achievements

- **Databricks Lakehouse Fundamentals** – Databricks (July 2026)
- **Oracle Certified Foundations Associate: Agentic AI** – Oracle (Aug 2026)
- Completed LeetCode SQL 50 study plan
- **3rd Best: "Code Craft Chase" Hackathon**, Team Cintel, School of Computing, SRM IST (Mar 2025); Selected Participant: Guidewire DevTrails and Study AI Launch Pad.
- Winner of 1st Prize in Football, Cricket, and Basketball at ISA Day, SRM; Cricket Team Captain (2026).